**Report:**

The approach for my code was to use recursive logic to complete the Towers of Hanoi problem. The main function “hanoi” is using the general registers r1, r2, and r2 to keep track of the source, destination, and auxiliary pegs respectively with r4 as temporary storage. The function checks the base case (no disks left to move) and performs recursion to move n-1 disks, handling the moving of the nth disk. The stack pointer “sp” is vital in the recursion process. Since ARM doesn’t automatically take care of call stacks for recursive functions, it is important to manually save and restore the state of the program. The stack stores the return address “lr” and the registers before making each call, which lets each function call be preserved and restored when it completes.

**Code:**